11. A decline in the sale of ice cream during November to March is associated with

(a) seasonal variation (b) cyclical variation

(c) irregular variation (d) secular trend

12. Business forecasts are made on the basis of

(a) future data (b) past data

(c) tax regulations (d) Government policies

13. The four components of time series in a multiplicative model are

(a) independent (b) interdependent

(c) constant (d) additive

14. In the least square theory the sum of squares of residuals is

(a) zero (b) minimum

(c) constant (d) maximum

15. No statistical techniques for measuring or isolating _____is available.

(a) cyclical variation (b) seasonal variation

(c) erratic fluctuations (d) secular trend

II. Give very short answers to the following questions

16. What is a time series?

17. What are the components of a time series?

18. Name different methods of estimating the trend?

19. Write short notes on irregular variation.

20. Mention the methods used to find seasonal indices?

21. What are the demerits of moving averages?

22. What are the merits of method of least squares?

23. Write the normal equations used in method of least squares?

24. Define forecasting.

25. What are the three types of forecasting?

26. What is a short-term forecast?

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